

# Project  
M.Sc. Statistics Lecture Notes in R Markdown

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## Objective

Your task is to convert the supplied Markdown textbook into a completely rewritten, pedagogically structured set of R Markdown lecture notes for an M.Sc. Statistics course.

The objective is **\*\*NOT\*\*** to reproduce the textbook.

The objective is to produce high-quality university lecture material that follows the syllabus exactly.

The final output should resemble a professionally written graduate-level textbook rather than converted notes.

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# Primary Source

The supplied Markdown file (document.md) has been generated from

"Statistical Computing with R"  
Maria L. Rizzo  
CRC Press

Use it only as a reference.

Never copy paragraphs verbatim.

Rewrite everything in your own words. Use 'StyleGuide.md'

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# Course

M.Sc. Statistics

Course:  
Statistical Computing using R

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# Unit

Unit 3

Random Number Generation and Simulation Techniques

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## # Syllabus

### ## 3.1 Theory of Random Number Generation

- Concept of pseudo-random numbers
- Desirable properties of random number generators
- Linear Congruential Generator (LCG)
- Multiplicative Congruential Generator (MCG)
- Mixed Congruential Generator
- Criteria for selection of random number generators

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### ## 3.2 Testing of Random Number Generators

#### Uniformity Tests

- Chi-square Goodness-of-Fit Test
- Kolmogorov-Smirnov Test
- Digit Frequency Test

#### Independence Tests

- Runs Test
- Gap Test
- Poker Test
- Serial Correlation Test

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### ## 3.3 Inverse Transformation Method

- Theory
- Quantile Function
- Properties
- Distribution of Quantile Function
- Algorithms
- Discrete distributions
- Continuous distributions

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### ## 3.4 Acceptance-Rejection Method

- Theory
- Conditional Distribution
- Choice of M
- Exponential Tilting
- Algorithms

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## ## 3.5 Generation Using Distributional Relationships

- Distributional Relationships
- Composition Method
- Convolution Method
- Mixture Distributions
- Chi-square
- t Distribution
- F Distribution

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## ## 3.6 Multivariate Random Variable Generation

- Bivariate distributions
- Multivariate distributions
- Conditional distributions

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### # Output Format

Each syllabus topic must become one independent R Markdown file.

Example

...

03-01\_Theory\_of\_Random\_Number\_Generation.Rmd

03-02\_Testing\_of\_Random\_Number\_Generators.Rmd

03-03\_Inverse\_Transformation\_Method.Rmd

03-04\_Acceptance\_Rejection\_Method.Rmd

03-05\_Distributional\_Relationships.Rmd

03-06\_Multivariate\_Random\_Variable\_Generation.Rmd

...

Each Rmd file must compile independently.

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### # YAML

Every file must begin with

```yaml

---

title:

author:

date: "`r Sys.Date()"`"

output:

```

html_document:
  toc: true
  toc_depth: 3
  number_sections: true
  code_folding: hide
  theme: cosmo
---
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# Writing Style

Write like an experienced university professor.

Audience:

M.Sc. Statistics students.

The writing should

- be mathematically rigorous
- be highly readable
- be explanatory
- avoid unnecessary jargon
- avoid AI-style writing
- avoid bullet-heavy writing
- explain intuition before mathematics

Each section should gradually move from intuition to formal theory.

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# Pedagogical Structure

Every R Markdown file should contain the following sections.

# Learning Objectives

Clearly state what students should learn.

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# Prerequisites

Mention concepts students should already know.

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# Introduction

Explain

- motivation

```

- historical background
- practical importance

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#### # Intuition

Develop intuition before introducing formal definitions.

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#### # Formal Definitions

Provide mathematically rigorous definitions.

Use LaTeX equations.

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#### # Theorems

Whenever applicable

State

Proof

Remarks

Examples

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#### # Mathematical Derivations

Show every important derivation step-by-step.

Never skip mathematical steps.

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#### # Algorithms

For every algorithm include

Purpose

Input

Output

Pseudo-code

Computational complexity

Flowchart description

Practical considerations

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# R Programming

Every concept must include R code.

Code must

compile

be executable

be commented

follow tidy style

avoid deprecated functions

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# Simulation Studies

Include simulations illustrating

- theory
- convergence
- properties
- limitations

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# Graphs

Generate figures using R.

Prefer

ggplot2

over base graphics unless demonstrating base R.

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# Numerical Examples

Every major theorem should have a worked numerical example.

Show

Problem

Solution

Interpretation

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# Applications

Discuss

Statistics

Machine Learning

Data Science

Finance

Reliability

Monte Carlo

Bayesian Statistics

Operations Research

Engineering

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# Advantages

Discuss strengths.

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# Limitations

Discuss weaknesses.

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# Practical Tips

Explain

common mistakes

implementation issues

numerical instability

precision problems

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# Summary

Summarize key concepts.

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# Key Formulae

Provide a formula sheet.

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# Key Terms

Provide glossary.

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# Exercises

Include

## Multiple Choice Questions

10 questions

Solutions

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## Short Answer Questions

10 questions

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## Long Answer Questions

10 questions

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## Programming Exercises

10 problems

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## Mini Projects

2 project ideas

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## # References

Provide references in APA format.

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## # Mathematical Notation

Use proper LaTeX.

Display equations whenever appropriate.

Number important equations.

Use aligned environments.

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## # Figures

All figures should be generated in R.

Never paste screenshots.

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## # Tables

Use `knitr::kable()`

or

`gt`

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## # Code Style

Every chunk should contain

```
```r
```

```
set.seed(123)
```

```
library(tidyverse)
```

```
```
```

Use chunk names.

Example

```
```r
```{r lcg-example}
```
```

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## # Teaching Philosophy

Always teach in the following order.

1. Motivation
2. Intuition
3. Definition
4. Theory
5. Mathematics
6. Algorithm
7. Example
8. R Implementation
9. Simulation
10. Interpretation
11. Applications
12. Exercises

Never change this order.

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## # Content Expansion

The textbook is only one source.

Whenever the syllabus requires more detail,  
expand using standard statistical knowledge.

Do not merely summarize the source.

Produce lecture notes.

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## # Random Number Generation

For all RNG algorithms include

History

Mathematics

Period

Seed

Parameter selection

Advantages

Disadvantages

Implementation

Simulation

Comparison with R

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# Code Quality

All R code must

compile

run without modification

contain comments

follow modern R practices

avoid deprecated packages

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# Important Rule

Never copy textbook paragraphs.

Rewrite everything.

Improve explanations.

Expand wherever necessary.

Always prioritize the syllabus over the textbook.

The final document should read like a graduate-level university textbook written specifically for this course.